CASE

Margaret, 79 · "I think I'm shrinking" · new mid-back kyphosis · denied trauma, recalls a hard sneeze · osteoporosis · **compression fractures at T7 and T9**

BIO 004 · WEEK 2 · CASE DEEP DIVE

The Vertebral Column & Thoracic Cage

An eighty-year-old who shrank three inches, a hard sneeze a month ago, and the geometry that made the wedge inevitable.

PATIENT CASE

Margaret is 79, brought to clinic by her daughter who says her mother has been complaining of a **mid-back ache for three weeks** and "**she looks shorter**." The chart from her last visit a year ago lists her at 5'5". Today the medical assistant measures her at **5'2"**.

On exam Margaret has a visible **thoracic kyphosis** ("dowager's hump") that her daughter says was not there at the last visit. The pain is in the **mid-back, worse with movement, better lying flat**. She **denies any fall or trauma**, but when pressed she recalls "**sneezing really hard about a month ago**." A DEXA scan two years prior showed osteoporosis. Today's lateral X-ray of the thoracic spine shows **anterior wedging of T7 and T9**.

The shape of the spine and the architecture of a typical vertebra tell you why a sneeze can fracture a bone in an osteoporotic seventy-nine-year-old, why the wedge always goes forward, and why the kyphosis appeared overnight.

> **GUIDING QUESTION**
>
> How does the vertebral column carry weight, what makes a typical vertebra strong in one direction and weak in another, and why does osteoporosis turn a sneeze into an injury that changes a person's height?

PREWORK

Companion to your Week 2 Vertebral Column & Thoracic Cage prework page. Terms, video, and spaced recall for the column, the curvatures, a typical vertebra, regional vertebrae, and the thoracic cage live there.

1 OVERVIEW

The column and its job

The vertebral column is a stack of **33 vertebrae** with intervertebral discs between most of them: *7 cervical, 12 thoracic, 5 lumbar, 5 sacral (fused into the sacrum), 4 coccygeal (fused into the coccyx)*. It protects the spinal cord, supports the head and trunk, anchors the ribs and pelvis, and absorbs and transmits load.

Two design problems the column had to solve: bear axial weight while still flexible, and protect a delicate cord while still allowing nerves to exit at every level. The column solves them by stacking bony rings (vertebrae) cushioned by fibrocartilage pads (discs), and by leaving lateral notches at each level for the spinal nerves.

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2 CURVATURES

Four curves that develop in order

The adult spine has four curves when viewed from the side. They are not random; they evolved to put the head over the pelvis efficiently and to spring-load the column for shock absorption.

Cervical lordosis

Concave posteriorly. Develops when the infant begins to hold the head up (~3 months). A *secondary* curvature.

Thoracic kyphosis

Convex posteriorly. Present from birth (*primary* curvature). Margaret's collapsed vertebrae are accentuating this normal curve into a pathologic exaggeration.

Lumbar lordosis

Concave posteriorly. Develops when the toddler begins to walk (~12 months). *Secondary*.

Sacral kyphosis

Convex posteriorly. Primary. Fused sacrum.

Why the wedge is anterior. Each vertebral body in the thoracic region bears the weight of everything above it slightly anterior to the center of the disc. Osteoporotic bone fails on the side under most compression. The body wedges forward. The wedge increases the kyphosis, which puts even more load on the next vertebra forward of center. The geometry feeds itself, which is why Margaret has TWO wedged vertebrae and why patients often develop a cascade.

3 TYPICAL VERTEBRA

Body in front, arch behind, processes everywhere

Most vertebrae share a common floor plan, with regional variations on the theme.

Body (centrum)
The anterior, weight-bearing portion. A cylindrical chunk of bone. This is what wedges in osteoporotic fracture.

Vertebral arch
The bony ring behind the body. Made of paired *pedicles* and *laminae*. Encloses the vertebral foramen.

Vertebral foramen
The hole through which the spinal cord passes. Stacked together they form the vertebral canal.

Spinous process
Projects posteriorly. What you palpate down someone's back.

Transverse processes (2)
Project laterally. Anchor muscles and ribs.

Articular processes (4)
Two superior, two inferior. Form facet joints with the vertebra above and below.

Intervertebral foramina
Lateral notches between adjacent vertebrae. Spinal nerves exit through these.

4 REGIONAL VERTEBRAE

Same theme, regional variations

REGION	NUMBER	DEFINING FEATURES
Cervical	7 (C1-C7)	small bodies; transverse foramina (vertebral artery); bifid spinous processes (C2-C6). *C1 (atlas)*: no body, ring shape, holds the skull. *C2 (axis)*: has the dens (odontoid process). *C7 (vertebra prominens)*: long palpable spinous process.
Thoracic	12 (T1-T12)	heart-shaped bodies; long, downward-pointing spinous processes; *costal facets* for rib articulation (on body and transverse processes). This is where Margaret's wedges occurred.
Lumbar	5 (L1-L5)	large kidney-shaped bodies built for weight; short, thick spinous processes; no costal facets.
Sacrum	5 fused (S1-S5)	triangular; articulates with the pelvic bones at the sacroiliac joints; carries the body's weight to the legs.
Coccyx	3-5 fused	vestigial tail; attachment for pelvic floor muscles.

Between adjacent vertebral bodies sit **intervertebral discs**: a tough outer ring of fibrocartilage (*annulus fibrosus*) around a gel-like core (*nucleus pulposus*). Discs absorb shock and allow modest motion. With age the nucleus pulposus loses water content; the disc thins, becomes stiffer, and contributes to height loss independent of vertebral fractures.

5 THORACIC CAGE

Twelve ribs, one sternum, and the floor that breathes

The thoracic cage is the bony container of the chest: **12 pairs of ribs** articulating posteriorly with thoracic vertebrae and anteriorly with the **sternum** (via costal cartilage), with the **diaphragm** as the floor.

True ribs (1-7)
Each connects directly to the sternum via its own costal cartilage.

False ribs (8-10)
Costal cartilages fuse and join the cartilage of rib 7 rather than the sternum directly.

Floating ribs (11-12)
No anterior attachment. End in the abdominal wall musculature.

Sternum
Three parts: manubrium, body, xiphoid process. The *sternal angle* (between manubrium and body) marks the level of the second rib and the T4/T5 disc.

Why the thoracic cage matters for Margaret: as her thoracic vertebrae wedge, the rib cage tips forward with the spine. The chest contour changes, the abdomen protrudes, and the diaphragm rides higher. Severe kyphosis can reduce lung volume by 10 to 15 percent.

✎ **DRAW ZONE 1** *-- Sketch the spine from the side. Show all four curves; label cervical lordosis, thoracic kyphosis, lumbar lordosis, sacral kyphosis. Mark which are primary vs secondary.*

Draw Margaret's exaggerated thoracic kyphosis next to a normal one for comparison.

✎ **DRAW ZONE 2** -- *Sketch a typical vertebra from above. Label body, vertebral foramen, pedicle, lamina, spinous process, transverse processes, superior and inferior articular processes.*

Then draw a wedged osteoporotic body next to a normal one. Show how the anterior collapse changes the disc angle.

RETURNING TO MARGARET

Why a sneeze fractured her spine

Margaret's case threads every concept on this page. Osteoporotic bone, anterior weight bearing, thoracic kyphosis geometry, intervertebral discs, regional vertebrae.

"*Three inches lost*"

Two wedged vertebrae account for some of it (each one removes ~1cm of anterior height). Additional disc thinning from age contributes the rest. Vertebral fractures plus dehydrated discs equal measurable height loss over years.

"*Why mid-back, not lumbar*"

Thoracic vertebrae bear weight slightly anterior to disc center, so the anterior cortex sees the most compression. Osteoporotic bone fails there first. Lumbar bodies are larger and more centrally loaded.

"*Why a sneeze did it*"

A hard sneeze generates significant axial force through the column (intra-abdominal pressure spikes plus rapid trunk flexion). In a vertebra with normal bone density it does nothing. In osteoporotic bone the anterior body cortex collapses.

"*Why the kyphosis appeared overnight*"

Two anteriorly wedged vertebrae tilt the upper torso forward. The change is visible the moment it happens. There is no gradual onset of an acute wedge fracture.

"*Risk of more fractures*"

A wedge fracture cascade is common: each fracture shifts load further anteriorly, accelerating fracture of the next vertebra forward of center. Treat the osteoporosis aggressively.

"*What this looks like on imaging*"

Lateral X-ray shows anterior wedging of the vertebral body. The posterior body height is preserved (no impingement on the cord). MRI can age the fracture: marrow edema means acute or subacute.

Body, arch, curve, load. When you go back to the prework, the typical vertebra diagram will land differently. Margaret's wedge is what happens when the bone in front gives out under a load the bone in back never has to carry.